



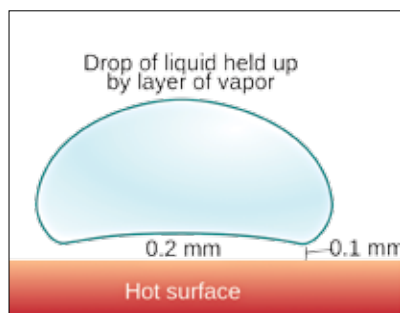
Living drilling ant

Study reveals extreme adaptations lets wood-boring ants carve complicated tunnel networks in their host trees. <https://digit.in/sep18-69-1>



Alcohol not safe at all

Drinking alcohol in moderation is more harmful than previously thought, according to a new study. <https://digit.in/sep18-69-2>



lead goes into evaporating the water and forms an insulating layer of steam around the fingers which protects them from burning. A similar trick is applied when showcasing a burning piece of camphor on one's hand or tongue, or even extinguishing a burning cigarette on one's hand or tongue.

TRICKS WITH PRODUCE

Trick: Coconut catches fire, opens to reveal blood inside

Science: Potassium permanganate, Sodium-water reaction and a soft eye on the coconut

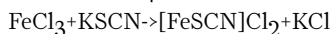
Explanation: This demonstration involves throwing water at a coconut until it catches fire, then cracking it open to reveal blood inside. To do this, first, a thick solution of potassium permanganate in water is created. The coconut has three "eyes" on the top out of which one is soft. The potassium permanganate solution is injected into the coconut through this soft eye. Then, a piece of sodium metal is placed in between the coconut fibres. Then, water is sprinkled on the coconut until it catches fire. This happens due to the highly exothermic (heat releasing) reaction between sodium and water that releases sodium hydroxide. Then, the coconut is cracked open to reveal the blood-like potassium permanganate solution with coconut water. Other tricks involving flowers inside a coconut and more are also done in a similar manner using the soft eye of the coconut. The sodium-water reaction is also used in other tricks to create fire, as most people don't expect throwing water to result in combustion. This is not some supernatural power, but just dressed up chemistry.

MAGIC HEALING WOUNDS

Trick: Blood from lemon (also magic wound)

Science: The reaction of a ferric solution with a thiocyanate compound

Explanation: This trick involves cutting of a lemon to show blood oozing out of the cut and the insides turning red as if blood is seeping in. A similar trick would be to show a knife touching one's palm and blood starts oozing out. Once the blood is wiped, the wound appears to have magically healed. This is done with the help of a reaction between a ferric solution and a thiocyanate compound. If you take an example:



An important result of this reaction is the intermediary $[\text{FeSCN}]^{+2}$ ion formed, which forms a blood red colour with a liquid. So, if a ferric solution is rubbed on to a surface, say, your palm or the surface of a lemon, and then a knife rubbed with a thiocyanate compound like potassium thiocyanate or ammonium thiocyanate is touched on it in a cutting action, it would appear to create a wound with blood where none exists.

FEATS WITH PHYSICS

Trick: A vessel appears to drink liquid/blood

Science: Creation of vacuum

Explanation: This trick makes it appear as if liquid, supposedly blood, is sucked into a vessel by a mysterious force. There are numerous ways to create fake blood that don't need to be elaborated on here. Now, this fake blood is poured onto a flat vessel. A small lamp is lit and placed at the centre and is covered by another vessel. Now, this flame gradually exhausts the available oxygen inside the second vessel, and leaves carbon-dioxide which is soluble in water. This CO_2 mixes with the water, leaving a vacuum behind which results in the liquid outside to be sucked into the inner vessel.

WALKING ON GLASS, NAILS

Trick: Walking, lying down on sharp objects - nails, glasses. Even additional force applied on top

Science: Distribution of force and the concept of pressure

Explanation: This one doesn't require as much trickery as it does faith in the laws of physics. Essentially, in many demonstrations, tricksters are seen lying down on a bed of nails or walking across a bed of broken glass without any cuts or harm to their skin. Some take this one step further by allowing weight to be put on themselves in this state. There's a pretty simple explanation behind this genre of tricks - pressure. High school physics teaches us that pressure equals force per unit area. While the sharpness of the glass and the nails might feel dangerous to an onlooker, it is not until the right pressure that is applied behind it that it becomes so. When a person is lying down on the bed of nails, their entire weight is distributed across the numerous nail tips, causing negligible pressure on the skin at each pin point as compared to the stepping on an



individual pin. Similarly, the bed of glass actually allows the glass to move about to adjust for the weight being put on top of it, distributing the weight of the person uniformly. In either case, care is taken to not rub against the sharp edge, which is why each step or act of lying down is done very slowly. In case of the glass, it is also made sure that pieces with particularly sharp edges are placed in parts of the bed where the trickster won't step on. 